

EE 115 Homework 5

Kaushik Vada

November 12, 2025

Problem 1 – Bandpass AM Chain

The message, modulator, and bandpass filter are

$$\begin{aligned} m(t) &= \cos(2\pi t) + 2\sin(4\pi t), \\ u(t) &= (m(t) + 3)\cos(200\pi t), \quad t \text{ is in ms,} \\ H(f) = H(-f) &= \begin{cases} \frac{f-98}{4}, & 98 \leq f \leq 102 \text{ kHz}, \\ 1, & 102 < f \leq 103 \text{ kHz}, \\ 0, & \text{otherwise for } f > 0, \end{cases} \end{aligned}$$

so the carrier is at $f_c = 100$ kHz and baseband frequencies are most conveniently described in kHz because t is expressed in milliseconds.

(a) Spectrum $M(f)$: Using the Fourier transform pairs for cosine and sine,

$$M(f) = \frac{1}{2}[\delta(f-1) + \delta(f+1)] + \frac{1}{j}[\delta(f-2) - \delta(f+2)], \quad f \text{ in kHz.}$$

Thus $M(f)$ consists of impulses at ± 1 kHz (height 0.5) from the cosine term and at ± 2 kHz (height 1 with odd symmetry) from the sine term.

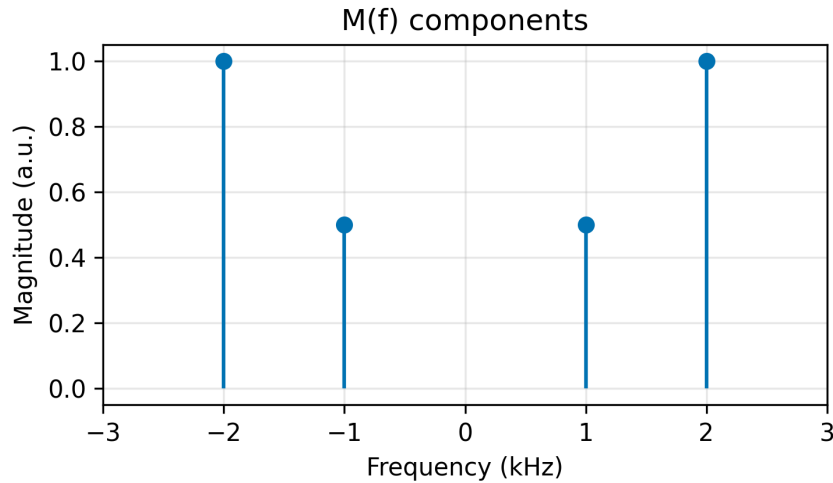


Figure 1: MATLAB magnitude spectrum $M(f)$ confirming impulses at ± 1 and ± 2 kHz.

- (b) **Spectrum $U(f)$:** Multiplication by $\cos(2\pi f_c t)$ produces the usual double-sideband translation,

$$U(f) = \frac{1}{2} \left[M(f - f_c) + M(f + f_c) \right] + \frac{3}{2} [\delta(f - f_c) + \delta(f + f_c)].$$

Consequently, positive-frequency lines appear at $f = \{98, 99, 100, 101, 102\}$ kHz with weights proportional to the baseband coefficients:

Component	f (kHz)	Time-domain amplitude
Carrier offset	100	3
Cosine sidebands	99, 101	0.5 each before filtering
Sine sidebands	98, 102	1 each before filtering

(Negative frequencies have the conjugate impulses, ensuring $u(t)$ is real.)

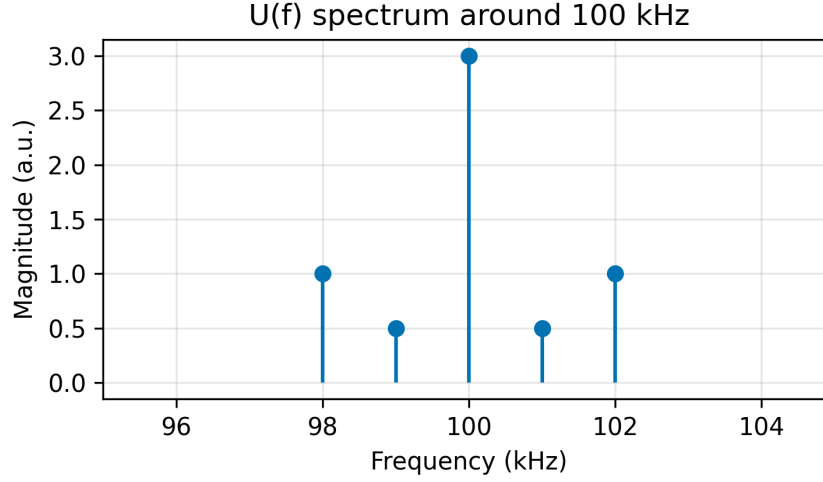


Figure 2: Simulated $|U(f)|$ showing the carrier and mirrored sidebands around 100 kHz.

- (c) **Spectrum $V(f)$:** $V(f) = H(f)U(f)$, so only those impulses that fall inside 98–103 kHz survive, each scaled by the gain $H(f)$ at that frequency. The surviving positive-frequency amplitudes are

Baseband freq.	f (kHz)	$H(f)$	Time amplitude in $v(t)$
Carrier (DC)	100	0.5	$1.5 \cos(2\pi f_c t)$
1 kHz upper	101	0.75	$0.375 \cos 2\pi(101\text{kHz})t$
1 kHz lower	99	0.25	$0.125 \cos 2\pi(99\text{kHz})t$
2 kHz upper	102	1	$\sin 2\pi(102\text{kHz})t$
2 kHz lower	98	0	suppressed

The negative-frequency impulses mirror these values.

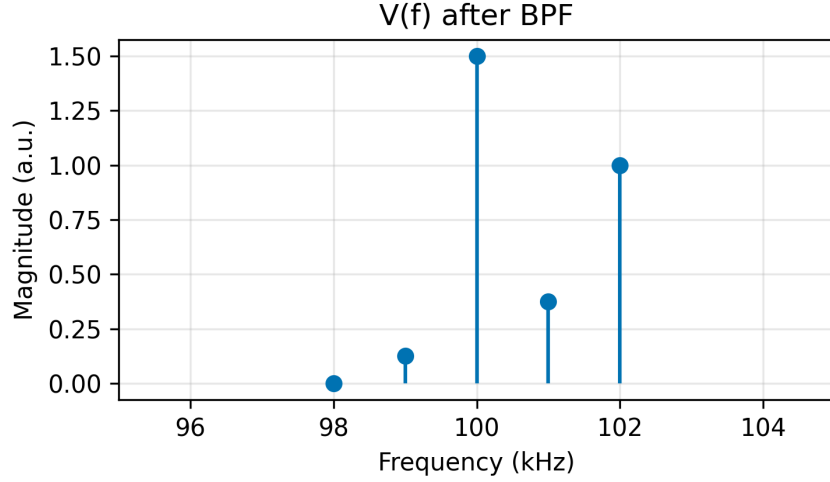


Figure 3: Magnitude $|V(f)|$ displaying the unequal sideband gains after the bandpass filter.

- (d) **Complex envelope of $h(t)$:** Let $\tilde{H}(f)$ denote the complex envelope (low-pass equivalent) of $h(t)$ with respect to f_c . It is obtained by retaining only the positive-frequency lobe around f_c and translating it to baseband:

$$\tilde{H}(f) = 2H(f_c + f) = \begin{cases} 0, & f < -2 \text{ kHz}, \\ 1 + \frac{f}{2}, & -2 \leq f \leq 2 \text{ kHz}, \\ 2, & 2 < f \leq 3 \text{ kHz}, \\ 0, & f > 3 \text{ kHz}. \end{cases}$$

Hence the complex-envelope magnitude is a ramp that rises from 0 at -2 kHz to 2 at 2 kHz, then stays at 2 until 3 kHz.

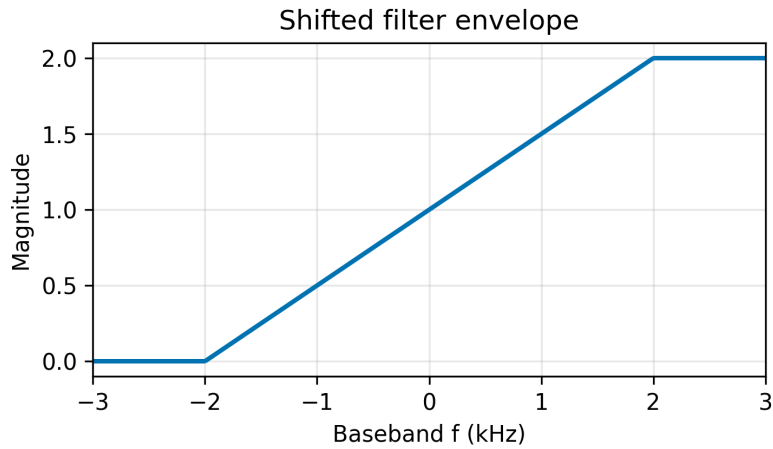


Figure 4: Baseband view of the filter magnitude $\tilde{H}(f)$ obtained by shifting the passband response down by 100 kHz.

(e) **Spectra of $v_c(t)$ and $v_s(t)$:** Writing $v(t) = v_c(t) \cos(200\pi t) - v_s(t) \sin(200\pi t)$ and using coherent detection,

$$\begin{aligned} v_c(t) &= 2 \text{LPF}\{v(t) \cos(200\pi t)\} = 1.5 + 0.5 \cos(2\pi t) + \sin(4\pi t), \\ v_s(t) &= -2 \text{LPF}\{v(t) \sin(200\pi t)\} = 0.25 \sin(2\pi t) - \cos(4\pi t), \end{aligned}$$

where the trigonometric arguments are in kHz because t is measured in ms. Their spectra therefore consist of impulses at DC, ± 1 kHz, and ± 2 kHz:

Signal	Line freq. (kHz)	Weight	Comment
$v_c(t)$	0	1.5	residual carrier
	± 1	$0.25 \delta(f \mp 1)$	cosine term
	± 2	$\pm \frac{1}{2j} \delta(f \mp 2)$	sine term
$v_s(t)$	± 1	$\pm \frac{0.25}{2j} \delta(f \mp 1)$	sine term
	± 2	$0.5 \delta(f \mp 2)$	cosine term

The unequal gains on the upper and lower sidebands force a nonzero quadrature component $v_s(t)$.

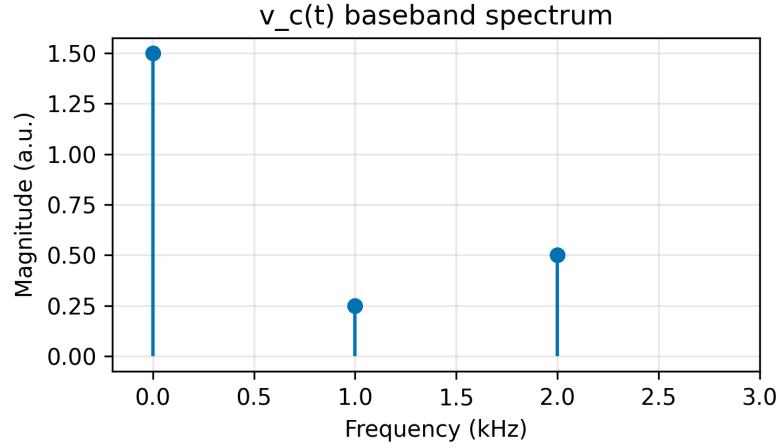


Figure 5: Spectrum of $v_c(t)$ emphasizing the DC bias and attenuated $\pm 1, \pm 2$ kHz lines.

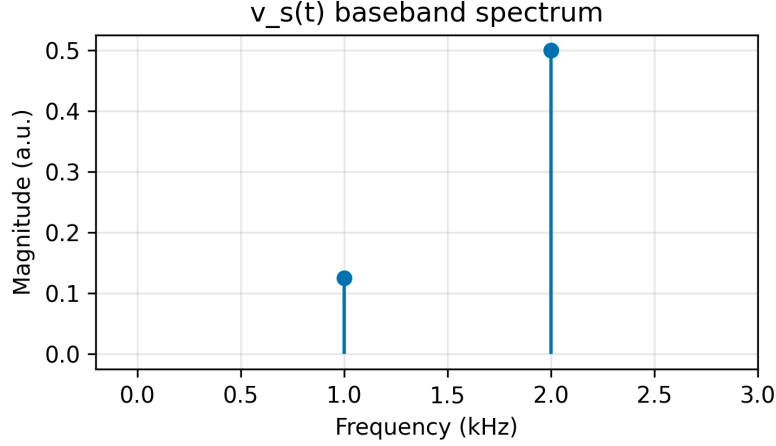


Figure 6: Spectrum of $v_s(t)$ showing the residual quadrature content generated by the asymmetric filter.

- (f) **Comparison of $v_c(t)$ with $m(t)$:** Both signals are baseband, but the BPF distorts the envelope:

$$m(t) = \cos(2\pi t) + 2\sin(4\pi t) \quad \Rightarrow \quad v_c(t) = 1.5 + 0.5\cos(2\pi t) + \sin(4\pi t).$$

The carrier leak introduces a DC offset of 1.5, the 1 kHz tone is attenuated by -6 dB, and the 2 kHz tone loses the lower sideband so its amplitude is halved. Thus $v_c(t)$ is a distorted (biased and unequalized) version of $m(t)$.

- (g) **Spectra of $h_c(t)$ and $h_s(t)$:** Using $h(t) = h_c(t)\cos(200\pi t) - h_s(t)\sin(200\pi t)$, the in-phase and quadrature frequency responses are

$$H_c(f) = H(f_c + f) + H(f_c - f), \quad H_s(f) = j[H(f_c - f) - H(f_c + f)],$$

with f expressed in kHz relative to baseband. For this filter:

- $H_c(f) = 1$ for $|f| \leq 3$ kHz and 0 elsewhere (a flat low-pass shape).
- $H_s(f)$ is odd and purely imaginary: it equals $+j$ on $-3 \leq f < -2$, ramps linearly as $-j f/2$ over $-2 \leq f \leq 2$, equals $-j$ on $2 < f \leq 3$, and is zero outside this band.

These sketches show that a purely real passband response still corresponds to coupled in-phase/quadrature low-pass filters whenever the upper and lower transition bands are asymmetric.

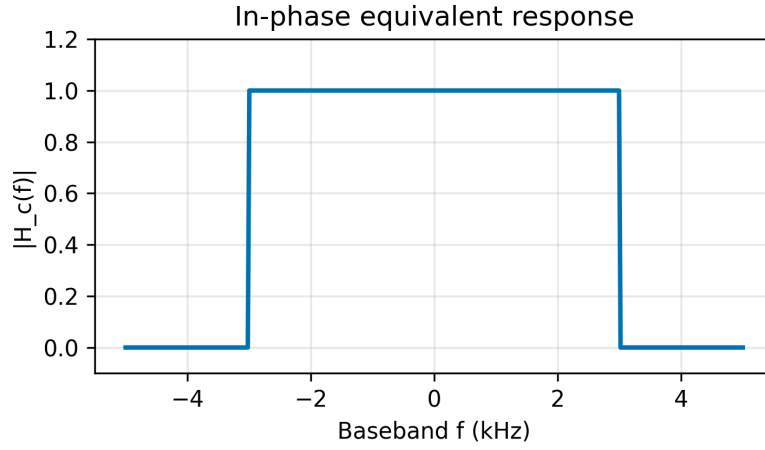


Figure 7: Numerical $|H_c(f)|$ confirming the unity-gain low-pass equivalent.

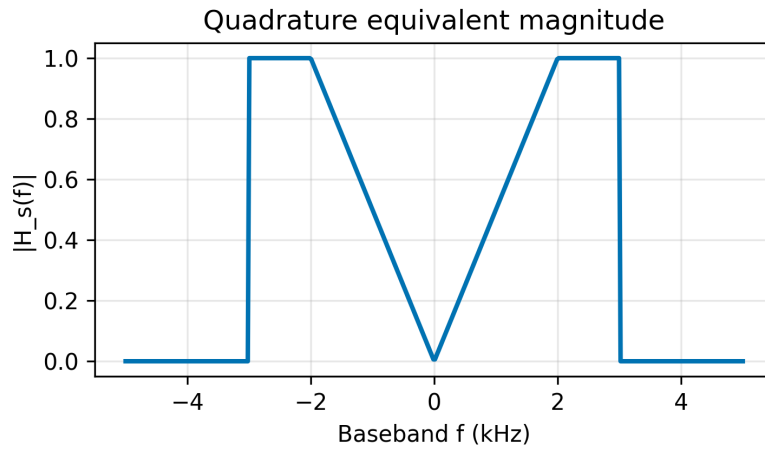


Figure 8: Magnitude of $H_s(f)$ highlighting the odd-symmetric quadrature response.

Problem 2 – RC Integrator Condition

For the RC network,

$$H(j\omega) = \frac{1}{j\omega RC} \frac{1}{1 + \frac{1}{j\omega RC}} = \frac{1}{1 + j\omega RC}.$$

To emulate the ideal integrator $\frac{1}{RC} \int_{-\infty}^t m(\tau) d\tau$, we need $H(j\omega) \approx \frac{1}{j\omega RC}$ over the entire message band. That happens when the magnitude of $\frac{1}{j\omega RC}$ is much smaller than 1, i.e.,

$$\left| \frac{1}{j\omega RC} \right| = \frac{1}{\omega RC} \ll 1 \quad \Rightarrow \quad \omega RC \gg 1.$$

Because the message spectrum is confined to $f \geq f_L$, the design rule is

$$2\pi f_L RC \gg 1 \quad \text{or equivalently} \quad f_L \gg \frac{1}{2\pi RC}.$$

Ensuring the lowest message frequency is at least a decade above $1/(2\pi RC)$ keeps the RC circuit within the 20 dB/dec roll-off region where it behaves like a scaled integrator.